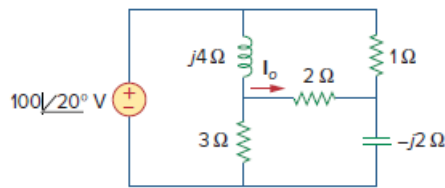


Problem

By nodal analysis, obtain current I_o in the circuit of Fig. 10.66.

Figure 10.66:



Step-by-step solution

Step 1 of 5

Refer to circuit diagram in Figure 10.66 in the textbook.

Draw the circuit with node notations.

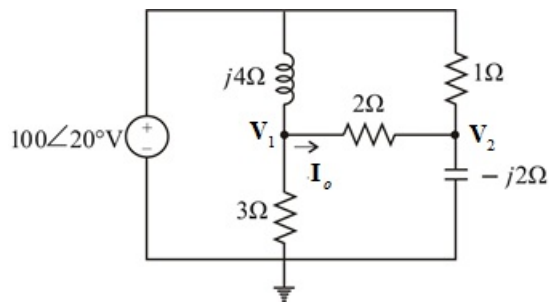


Figure 1

Step 2 of 5

From the circuit,

$$I_o = \frac{V_1 - V_2}{2} \dots\dots (1)$$

Apply Kirchhoff's current law at node V_1 .

$$\frac{V_1 - 100\angle 20^\circ}{j4} + \frac{V_1 - V_2}{2} + \frac{V_1}{3} = 0$$

$$\frac{3(V_1 - 100\angle 20^\circ) + j6(V_1 - V_2) + j4V_1}{j12} = 0$$

$$3V_1 + j6V_1 - j6V_2 + j4V_1 = 300\angle 20^\circ$$

$$(3 + j10)V_1 - j6V_2 = 300\angle 20^\circ$$

$$(3 + j10)V_1 = 300\angle 20^\circ + j6V_2$$

$$V_1 = 17.17 - j23.04 + (0.55 + j0.165)V_2 \dots\dots (2)$$

Step 3 of 5

Apply Kirchhoff's current law at node V_2 .

$$\frac{V_2 - V_1}{2} + \frac{V_2}{-j2} + \frac{V_2 - 100\angle 20^\circ}{1} = 0$$

$$\frac{V_2 - V_1}{2} + \frac{jV_2}{2} + \frac{V_2 - 100\angle 20^\circ}{1} = 0$$

$$\frac{V_2 - V_1 + jV_2 + (2)(V_2 - 100\angle 20^\circ)}{2} = 0$$

$$V_1 - (3 + j)V_2 = -200\angle 20^\circ$$

Substitute $17.17 - j23.04 + (0.55 + j0.165)V_2$ for V_1 .

$$17.17 - j23.04 + (0.55 + j0.165)V_2 - (3 + j)V_2 = -200\angle 20^\circ$$

$$(0.55 + j0.165 - 3 - j)V_2 = -187.94 - j68.4 - 17.17 + j23.04$$

$$(-2.45 - j0.835)V_2 = -205.11 - j45.364$$

$$V_2 = \frac{-205.11 - j45.364}{-2.45 - j0.835}$$

$$= 80.66 - j8.974$$

$$= 81.157\angle -6.348^\circ \text{ V}$$

Step 4 of 5

Recall equation (2).

$$V_1 = 17.17 - j23.04 + (0.55 + j0.165)V_2$$

Substitute $81.157\angle -6.348^\circ \text{ V}$ for V_2 .

$$V_1 = 17.17 - j23.04 + (0.55 + j0.165)(81.157\angle -6.348^\circ)$$

$$= 17.17 - j23.04 + 45.844 + j8.373$$

$$= 63.0136 - j14.67$$

$$= 64.698\angle -13.1^\circ \text{ V}$$

Step 5 of 5

Calculate the output current.

$$I_o = \frac{V_1 - V_2}{2}$$

Substitute $64.698\angle -13.1^\circ \text{ V}$ for V_1 and $81.157\angle -6.348^\circ \text{ V}$ for V_2 .

$$I_o = \frac{64.698\angle -13.1^\circ - 81.157\angle -6.348^\circ}{2}$$

$$= \frac{-17.645 - j5.694}{2}$$

$$= -8.823 - j2.847$$

$$= 9.25\angle -162.12^\circ \text{ A}$$

Therefore, the output current, I_o is $9.25\angle -162.12^\circ \text{ A}$.